Lecture 6.3: Machine Learning in Economic Research

AREC 570 / ECON 530 · Unit 6

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Before class

- Athey and Imbens, “The State of Applied Econometrics: Causality and Policy Evaluation” (Athey and Imbens 2017)
- Athey and Imbens, “Machine Learning Methods That Economists Should Know About” (Athey and Imbens 2019)
- Mullainathan and Spiess, “Machine Learning: An Applied Econometric Approach” (Mullainathan and Spiess 2017)

Readings and course handouts are available on Canvas.

Example papers in this lecture

- Background: Einav and Levin (2014); Bahoo et al. (2025)
- Overview for agribusiness: Maruejols et al. (2025)
- Prediction: Zhao et al. (2025)
- Prediction within a larger analysis: Li and Yu (2025); Belloni et al. (2014)
- Heterogeneous treatment effects: Stetter et al. (2022)
- Further reading: Bittmann's guide from OLS to causal machine learning (Bittmann 2025)

Machine learning in a nutshell

If you take away one idea today: ML is a prediction method learns from observations we already have, then we check how well it predicts observations it has not seen

1. Choose the outcome to predict and the information available when a prediction is made
2. Fit a prediction rule using one group of observations
3. Compare its predictions with actual outcomes in a separate group
4. Report how much better it performs than a simple benchmark

In a causal study, machine learning can help account for differences between groups before estimating an effect

How does Facebook recognize faces in photos?

Algorithms use a large dataset of photos labeled as having a face or not to estimate a function $f(x)$ that predicts the presence y of a face from pixels x.

This similarity to econometrics raises questions:

- Are these algorithms merely applying standard techniques to novel and large datasets?
- If there are fundamentally new empirical tools, how do they fit with what we know?
- As empirical economists, how can we use them?

Source: Mullainathan and Spiess (2017), page 87.

Different goals: prediction and parameter estimation

Many economic applications estimate a parameter, such as a treatment effect or price sensitivity

Causal inference in a RF sense asks us to predict outcomes under conditions that were not observed, trying to replicate the gold standard of a RCT

Supervised machine learning methods are often designed to predict an outcome accurately for new observations

Note: Economists can also use prediction methods within procedures that estimate effects

This lecture covers where these methods came from, the three roles they play in empirical economics, and how a methods section reports them

Source: Mullainathan and Spiess (2017), pages 87 to 88.

Evaluating predictions and parameters

	Prediction, $\widehat{y}$	**Parameter estimation, $\widehat{\beta}$**
Target	The outcome for a new observation	An effect or behavioral parameter
Common techniques	LASSO, random forests, and other prediction methods	OLS, IV, difference in differences, and structural models
Evaluation	Accuracy on data not used in estimation	Identification: the source of variation and its assumptions
Verifiability	Directly testable: predictions and outcomes are both observed	Rests on assumptions about the data-generating process

Source: Mullainathan and Spiess (2017), pages 88 and 92.

Background

Machine learning's use in applied economics grew over the past 20 years

Period	Development
1984 to 2001	Regression trees (Breiman et al., 1984), the LASSO (Tibshirani, 1996), and random forests (Breiman, 2001)
1990s	Neural networks studied in econometrics, but they "did not catch on at the time"
2001	Breiman's "two cultures": statistics relied on data models, while an algorithmic culture grew in other fields
2014	Early reviews for economists: Varian; Einav and Levin; Belloni, Chernozhukov, and Hansen
2017 to 2019	Mullainathan and Spiess; causal forests (Wager and Athey); Athey and Imbens's review for economists
2020 to 2025	Reviews and special issues in agricultural and applied economics: Storm et al. (2020); Maruejols et al. (2025)

Sources: Athey and Imbens (2019), pages 686 to 688, 696 to 698, and 701; Mullainathan and Spiess (2017), page 87; Storm et al. (2020); Maruejols et al. (2025).

Why economists adopted machine learning

1. **More readily available access to “big data”**
    - Properties: many units, many variables per unit, and unstructured sources such as satellite images and text
    - Challenge organizing and studying this type of data (or even to decide which variables to construct) without a simplifying conceptual framework
    - Better data allow for (1) sharper tests of existing models, (2) tests of theories that had previously been difficult to assess, and (3) assistance choosing controls or functional forms
2. **Prediction questions**
    - Some policy problems require an accurate forecast and no causal parameter

Sources: Athey and Imbens (2019), pages 686 to 687; Mullainathan and Spiess (2017), pages 88, 98, and 102, Einav and Levin (2014)

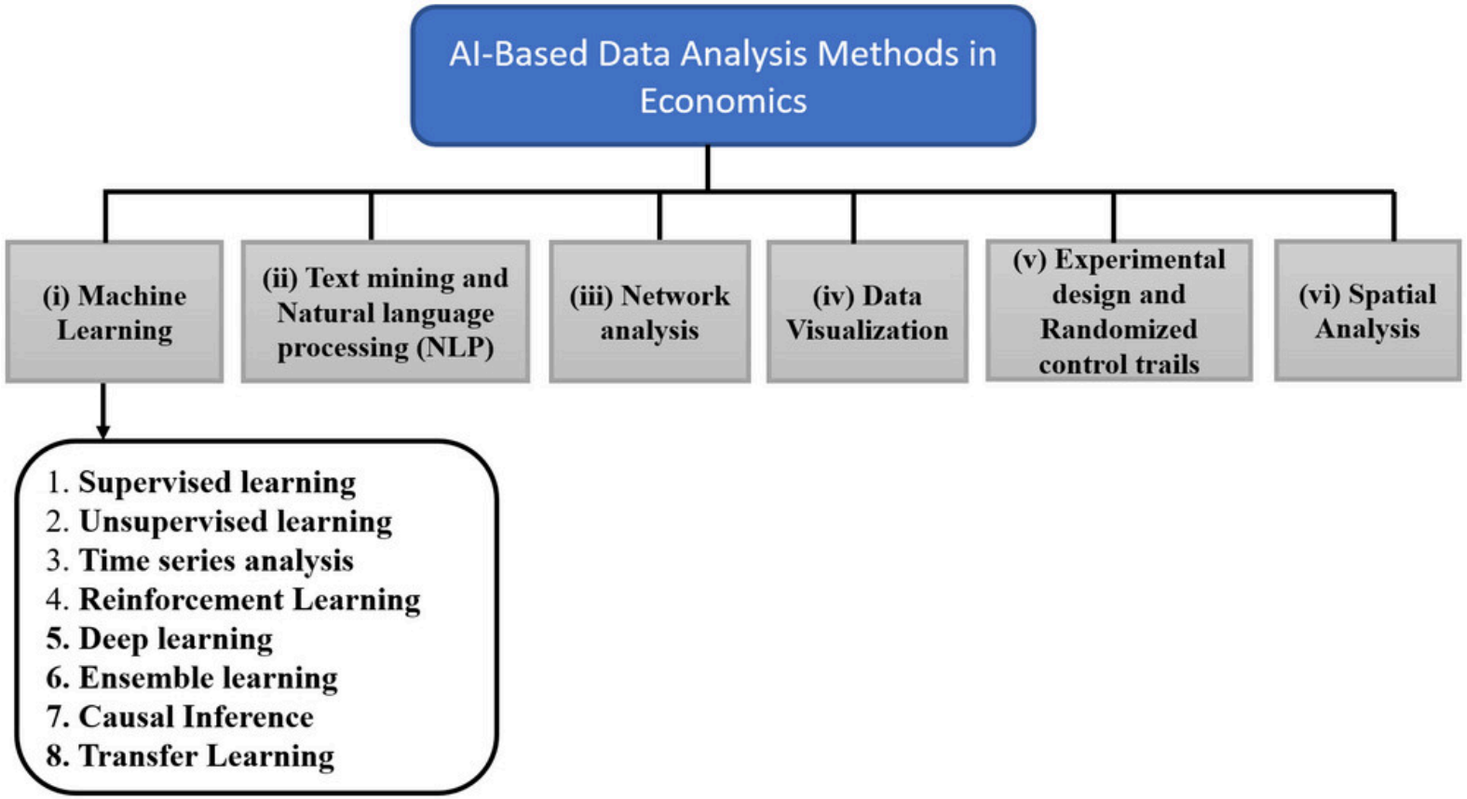
Why adoption was slow

Economics journals emphasize formal properties of estimators, such as consistency and asymptotic normality, which early machine learning methods did not provide

However, recent work established inference for adapted methods, including double selection and causal forests

Sources: Athey and Imbens (2019), pages 686 to 687; Mullainathan and Spiess (2017), pages 88, 98, and 102.

Machine learning within the AI-based tool landscape



Source: Bahoo et al. (2025), Section 4.3 and Figure 6.

How machine learning works

How a house-price prediction rule learns

Imagine we have recent home sales and want to predict the sale price of another house

1. For each sold house, record its sale price and characteristics such as size and location
2. Set aside some houses before fitting a model
3. Use the remaining houses to fit a rule that turns characteristics into predicted prices
4. Measure each mistake by comparing a prediction with the actual sale price

For a house that sold for **$300,000**, a prediction of **$320,000** misses by **$20,000**

Is this rule sufficient, or should it be adjusted to reduce the average mistake on the houses used for fitting?

Source: Mullainathan and Spiess (2017), page 89

How do we score a house-price prediction?

Suppose a house sold for **$300,000** and our model predicted **$320,000**.

Loss function	Score for this house	What it emphasizes
Absolute error	$\|320 - 300\| = 20$	The size of each mistake
Squared error	$(320 - 300)^2 = 400$	Larger mistakes receive more weight

A **loss function** measures how wrong a prediction is for one observation.

For squared error:

$$L(\widehat{f}(x_i), y_i) = (y_i - \widehat{f}(x_i))^2$$

Here, y_i is the actual price and $\widehat{f}(x_i)$ is the predicted price.

Source: Mullainathan and Spiess (2017), page 89

What are we trying to minimize?

We want a prediction rule that makes the smallest mistakes, on average, for **new houses**.

This is called **expected loss**:

$$E_{(Y,X)}[L(\widehat{f}(X), Y)]$$

For squared error:

$$E_{(Y,X)}[(Y - \widehat{f}(X))^2]$$

- L measures the loss for one house.
- E takes the average loss across the population.
- $\widehat{f}(X)$ predicts price using house characteristics.

The goal: Find the prediction rule $\widehat{f}$ that minimizes expected loss.

Source: Mullainathan and Spiess (2017), page 89

How do we evaluate a prediction rule?

We cannot directly calculate expected loss because we do not observe every possible future house.

Instead, we use the data we have:

1. **Training data:** Choose a prediction rule that minimizes average loss across observed houses. This is called *empirical loss*.

$$\frac{1}{n}\sum_{i=1}^{n} L(\widehat{f}(x_i), y_i)$$

Source: Mullainathan and Spiess (2017), pages 89 to 90

Testing the prediction rule

2. **Test data:** Apply the fitted rule to houses that were set aside and not used to train the model.
3. **Evaluate:** Calculate average loss on the test houses to estimate how well the rule predicts new observations.

Why? A rule that performs well on training data may not predict well on new data.

The next paper reports held-out R^2 as a summary of predictive fit. Higher held-out R^2 means better predictions relative to predicting the same average for every house

Source: Mullainathan and Spiess (2017), pages 89 to 90

An example: predicting house values with OLS and machine learning

Table 1

Performance of Different Algorithms in Predicting House Values

Method	*Prediction performance (R^2)*		*Relative improvement over ordinary least squares by quintile of house value*				
	Training sample	*Hold-out sample*	1st	2nd	3rd	4th	5th
Ordinary least squares	47.3%	41.7% [39.7%, 43.7%]	–	–	–	–	–
Regression tree tuned by depth	39.6%	34.5% [32.6%, 36.5%]	−11.5%	10.8%	6.4%	−14.6%	−31.8%
LASSO	46.0%	43.3% [41.5%, 45.2%]	1.3%	11.9%	13.1%	10.1%	−1.9%
Random forest	85.1%	45.5% [43.6%, 47.5%]	3.5%	23.6%	27.0%	17.8%	−0.5%
Ensemble	80.4%	45.9% [44.0%, 47.9%]	4.5%	16.0%	17.9%	14.2%	7.6%

Source: Mullainathan and Spiess (2017), pages 89 to 90 and Table 1.

Using data on **51,808 homes** from the American Housing Survey:

1. Fit OLS and several machine learning models using 10,000 homes and 150 characteristics
2. Predict values for the remaining 41,808 homes
3. Compare prediction accuracy using held-out R^2

Two findings:

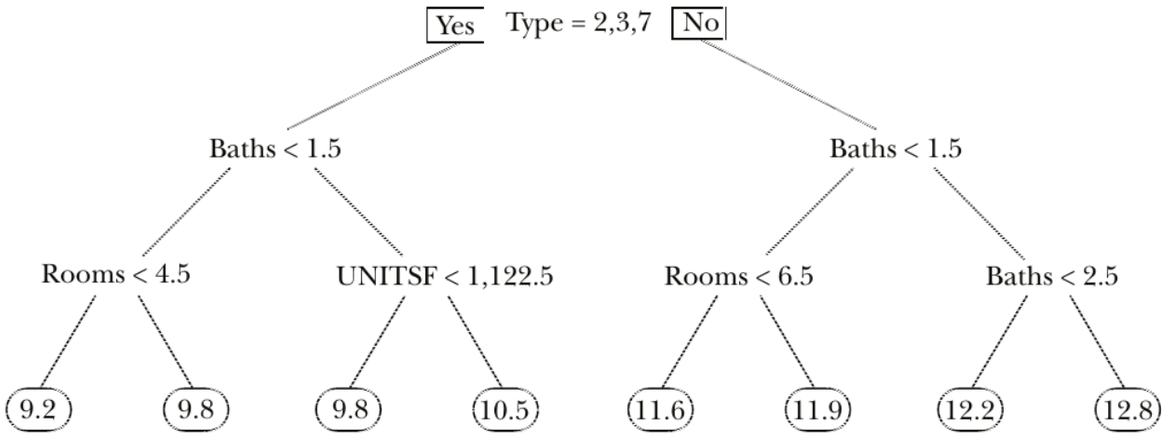
- Random forest: 45.5% held-out R^2, compared with 41.7% for OLS
- Random forest: 85.1% training R^2, but only 45.5% held-out R^2

How does a regression tree predict house values?

Unlike OLS, a regression tree does not require us to specify a linear relationship. Instead, it divides houses into groups with similar values.

Figure 1

A Shallow Regression Tree Predicting House Values



Note: Based on a sample from the 2011 American Housing Survey metropolitan survey. House-value predictions are in log dollars.

1. Start with all houses in one group
2. Find the yes-or-no split that produces the greatest reduction in squared prediction error (here, house type)
3. Repeat within each new group, using characteristics such as bathrooms, rooms, and square footage
4. Predict the value of a house using the average value of houses in its final group (a *leaf*)

Source: Mullainathan and Spiess (2017), pages 90 to 91 and Figure 1.

Why not keep dividing the tree?

Each additional split allows a regression tree to fit the training houses more closely.

1. More splits create smaller groups of similar houses
2. Smaller groups can reduce prediction errors in the training data
3. But very specific groups may capture patterns unique to those houses
4. Those patterns may not hold for new observations

Overfitting occurs when a model fits the training data closely but predicts poorly on new data.

Regularization limits model flexibility to improve out-of-sample prediction.

Source: Mullainathan and Spiess (2017), pages 91 to 93 and Table 2.

How do we choose the right tree depth?

A deeper tree may fit the training data better, but how deep should it be?

Cross-validation evaluates different model settings using only the training data.

1. Divide the training data into several groups (*folds*)
2. Fit the model using all but one fold
3. Predict outcomes for the excluded fold and calculate prediction error
4. Repeat until every fold has been excluded once
5. Choose the tree depth with the lowest average prediction error

Then fit the selected model on all training data and evaluate it on the **held-out test data**.

Sources: Mullainathan and Spiess (2017), pages 92 to 93; Bittmann (2025), Section 3.

How do the methods differ?

Each method learns a prediction rule from training data, but differs in how it generates predictions and controls complexity.

Method	How it predicts	How it controls flexibility
OLS	Estimates a linear combination of characteristics	Researcher specifies variables and interactions
Regression tree	Divides observations into groups	Limits on tree depth and group size

Sources: Mullainathan and Spiess (2017), pages 92 to 94 and Table 2; Athey and Imbens (2019), pages 690 and 695 to 699.

How do the methods differ? (continued)

Method	How it predicts	How it controls flexibility
LASSO	Estimates a linear combination of characteristics	Penalizes coefficients, shrinking some to zero
Random forest	Averages predictions from many trees	Randomization, averaging, and tree settings

Common goal: Minimize prediction error on new, unseen observations.

Sources: Mullainathan and Spiess (2017), pages 92 to 94 and Table 2; Athey and Imbens (2019), pages 690 and 695 to 699.

Examples of how machine learning is used in empirical economics

1. Prediction as the objective: Predicts an outcome, such as household food insecurity

2. Prediction within a larger analysis: Selects the variables that enter an economic model or a treatment-effect regression

3. Heterogeneous treatment effects: Estimates how a treatment effect varies across farms or households

Sources: Mullainathan and Spiess (2017); Athey and Imbens (2019), pages 686 to 688; Maruejols et al. (2025).

Role 1: Prediction as the objective

Zhao et al. (2025): predicting calorie poverty

Research question: Can household and market characteristics help identify households at risk of inadequate calorie intake?

1. Use three days of food records to measure calorie intake for 1,080 rural households
2. Classify each household as calorie poor or not calorie poor
3. Use household, village, and food-market characteristics to predict that classification
4. Fit a random forest, LASSO, and logit model using 756 households
5. Evaluate predictions on the remaining 324 households

Unlike house values, the outcome is binary: a household is either classified as calorie poor or not.

Source: Zhao et al. (2025), Sections 3 to 5.

Does machine learning improve classification?

Model	Test-sample accuracy
Random forest	65.7%
LASSO	61.1%
Logit benchmark	58.9%

Accuracy measures the percentage of households classified correctly.

The random forest improves accuracy by **6.8 percentage points** relative to logit.

But accuracy does not tell us which households were misclassified.

- **False positive:** A household is predicted to be calorie poor when it is not
- **False negative:** A calorie-poor household is not identified

Source: Zhao et al. (2025), Sections 3 to 5 and Table 3.

Do important predictors identify causes?

Zhao et al. find that household income and access to food markets help predict calorie poverty.

But a variable that improves prediction is not necessarily a cause of the outcome.

- A predictor may be correlated with other factors that affect calorie poverty
- Highly correlated predictors may provide similar information
- LASSO may select one predictor and exclude another, depending on the sample

Predictor importance tells us what helps classify households, not what would happen if we changed that characteristic.

Sources: Zhao et al. (2025), Section 5.2; Mullainathan and Spiess (2017), page 97, Figure 2; Stetter et al. (2022), page 740, footnote 17.

Role 2: Prediction within a larger analysis

Li and Yu (2025): selecting apple attributes

Research question: How much are consumers willing to pay for apple attributes such as origin, size, and production method?

1. Ask 100 consumers to choose between apples with different attributes and a no-purchase option
2. Use LASSO to identify which attributes help predict consumer choices
3. Estimate a conditional logit model using the selected attributes
4. Calculate willingness to pay from the estimated coefficients

Role of ML: Select variables to include in an economic model.

Source: Li and Yu (2025), Sections 2 and 3.

How does LASSO select attributes?

LASSO estimates coefficients while penalizing their absolute magnitude.

As the penalty increases, some coefficients shrink to zero.

In Li and Yu (2025):

- Tenfold cross-validation determines the penalty
- **Organic** and **Japan** have coefficients of zero at the selected penalty
- **Price** remains because the authors exclude it from the penalty

The excluded attributes provide little additional predictive information in this fitted model, given the other attributes.

Does a coefficient of zero mean consumers do not value the attribute?

Source: Li and Yu (2025), Section 3.1, Figure 2, and Table 4.

LASSO changes willingness-to-pay estimates

Estimated willingness to pay relative to China, in euros per kg:

Apple attribute	All attributes	After LASSO selection
France	1.20	1.53
Italy	1.42	1.40
Small apples	1.30	1.41
Organic	−0.04	Excluded

Source: Li and Yu (2025), Section 3, Tables 6 and 7.

What changes when LASSO selects attributes?

After removing the selected attributes, estimated willingness to pay changes for some of the remaining characteristics.

LASSO selects variables based on their contribution to the model's predictive objective, subject to its penalty.

But the variables that produce the best predictions are not necessarily the variables we need to estimate a particular economic parameter.

Source: Li and Yu (2025), Section 3, Tables 6 and 7.

Role 3: Heterogeneous treatment effects

How does prediction help estimate an effect?

Suppose an agri-environment program encourages farmers to reduce fertiliser use.

The research question is: Do programs have the same effect on every farm?

The average treatment effect tells us how participation changes fertiliser use across farms.

But the effect may differ depending on:

- Farm size
- Location
- Crop production and yield potential
- Other farm characteristics

Sources: Athey and Imbens (2019), pages 701 to 704; Stetter et al. (2022), Sections 2.3 and 4.1 to 4.3.

Why estimate heterogeneous treatment effects?

The **Average Treatment Effect (ATE)** measures the average effect of participation across all farms:

$$ATE = E[Y(1) - Y(0)]$$

But the program may affect farms differently depending on their size, location, or production practices.

The **Conditional Average Treatment Effect (CATE)** measures the average effect for farms with particular characteristics:

$$\tau(x) = E[Y(1) - Y(0) \mid X = x]$$

ML application: Causal forests can identify patterns in treatment-effect heterogeneity without requiring researchers to specify every interaction in advance.

How does a causal forest differ from a regression tree?

A regression tree divides observations into groups to predict an **outcome**.

A causal forest seeks groups with different **treatment effects**.

1. Predict treatment participation using observed characteristics
2. Predict the outcome using observed characteristics
3. Account for those predicted relationships when comparing treated and untreated farms
4. Use many trees to estimate how treatment effects vary across farm characteristics

The goal: Estimate treatment-effect heterogeneity rather than minimize ordinary outcome prediction error.

Sources: Athey and Imbens (2019), pages 701 to 704; Stetter et al. (2022), Sections 2.3 and 4.1 to 4.3.

Stetter et al. (2022): do program effects differ across farms?

Research question: Does participation in an agri-environment scheme change farm outcomes, and do those effects vary across farms?

1. Observe 2,758 Bavarian farms in 2014, including 1,641 program participants
2. Collect farm characteristics, program participation, and environmental outcomes
3. Use machine learning to account for observed differences between participating and nonparticipating farms
4. Estimate heterogeneous treatment effects using a causal forest

The authors examine four outcomes, including fertiliser expenditure, pesticide expenditure, land-use diversity, and emissions.

Source: Stetter et al. (2022), Sections 2.3, 4.1, and 4.2.

Which farms have detectable program effects?

Outcome	Share of farms with a statistically detectable effect
Fertiliser expenditure	33% show reductions
Land-use diversity	55% show increases
Pesticide expenditure	7% show improvements
Emissions	Under 1% show reductions; about 4% show increases

Among farms with detectable reductions in fertiliser expenditure, the estimated reduction averages **€14 per hectare**. Estimated effects vary by location, farm size, and yield potential. **Policy relevance:** Differences in estimated effects may inform how program payments are targeted.

Source: Stetter et al. (2022), Sections 5.2, 5.3, and 6 and Table 3.

What makes a causal forest estimate credible?

A causal forest can estimate differences in treatment effects, but **machine learning alone does not establish causality**.

The analysis relies on two important assumptions:

1. **Selection on observables:** After accounting for recorded characteristics, treated and untreated farms are comparable in their potential outcomes
2. **Overlap:** Farms with similar characteristics have a positive probability of participating or not participating

Stetter et al. exclude 23 farms that lack suitable comparisons and examine sensitivity to unobserved confounding.

The distinction: Machine learning helps estimate relationships; causal identification determines whether they can be interpreted as treatment effects.

Source: Stetter et al. (2022), Sections 4.3 and 5.1.

What role does machine learning play in economic research?

Role	Research objective	Contribution of ML
Prediction	Identify households at risk of calorie poverty	Improve classification accuracy
Model selection and estimation	Estimate preferences or a coefficient of interest	Select variables and controls
Heterogeneous treatment effects	Estimate how program impacts differ across farms	Flexibly estimate differences in treatment effects

Sources: Mullainathan and Spiess (2017); Athey and Imbens (2019); Zhao et al. (2025); Li and Yu (2025); Belloni et al. (2014); Stetter et al. (2022).

Three questions to ask when reading a paper

1. What is the quantity the researchers want to estimate?
2. What does machine learning contribute to that objective?
3. What assumptions are needed to interpret the results?

Sources: Mullainathan and Spiess (2017); Athey and Imbens (2019); Zhao et al. (2025); Li and Yu (2025); Belloni et al. (2014); Stetter et al. (2022).

Reporting a machine learning method

What to report in a machine learning methods section

Item	Example
1. What the method does in the study	Zhao et al. use it to predict calorie poverty
2. What is predicted and which information is used	Calorie poverty from household and market characteristics
3. Which observations are used for fitting and testing	70% training, 30% test
4. How method settings are chosen	Cross-validation; number of trees
5. How well it performs on test observations	Classification accuracy, compared with logit
6. If effects are claimed, what supports the causal comparison	Stetter et al. discuss comparable farms and overlap
7. How another researcher could repeat the analysis	Software, package versions, random seed, and code

Sources: Zhao et al. (2025); Li and Yu (2025); Stetter et al. (2022); Athey and Imbens (2019), pages 686 to 687.

What to consider for your proposal

- **The role:** does the question concern predicting an outcome, selecting variables, or how an effect varies?
- **The claim:** does the conclusion require a causal effect? If so, what is the design, and where does machine learning enter it?
- **The benchmark:** which simpler model or rule should the method outperform?
- **The evidence:** which out-of-sample measure, or which identification check, will the paper report?

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